DENSO HACKATHON · AI CHALLENGE

# Accumulate Master Know-How for AI to Learn

Digitizing expert craftsman know-how on the factory floor — capturing tacit knowledge directly from operational workflow and verbal reasoning without manual documentation.

Context → The Challenge → Reference Case → Solution

Team: **MentorMind** October 2026 Process: Kitting & Packing Assembly

— CONTEXT · BỐI CẢNH

# Expert Knowledge Is Leaving the Factory Floor

Tri thức chuyên gia đang rời khỏi nhà máy — khoảng cách lớn giữa tài liệu lý thuyết và thao tác thực tế tại dây chuyền.

### Master Artisans Retiring

Veteran technicians retire, taking with them decades of unwritten, tacit know-how accumulated through hands-on craftsmanship.

### SOPs Cannot Keep Up

Bulky manual SOPs are slow to update and fail to capture micro-gestures, subtle hand orientations, and live troubleshooting instincts.

### Multimodal VLM + RAG Opportunity

Vision-Language Models and GraphRAG present a historic opportunity to learn and replicate this knowledge — if captured ethically and timely.

DENSO CONTEXT Automotive Precision Manufacturing

## High Precision & Craftsmanship Dependence

Denso automotive components require ultra-tight tolerances. Kitting and packing processes heavily depend on technician tactile feedback:

"A slight deviation in placement angle or insertion pressure can trigger cascading defects across an entire production lot."

**80%**
Tacit unwritten knowledge

**≤ 5 Mins**
Defect containment window

— THE CHALLENGE · ĐỀ BÀI

# Accumulate Master Know-How for AI to Learn

Capture expert workflows and reasoning into a centralized knowledge base, leveraging LLMs to reuse tacit wisdom for similar tasks — with zero manual documentation.

DELIVERABLE 01

## Standardized Capture Template & Process

Standardized WorkProfile JSON Schema capturing granular steps, key tactile points, underlying reasons ("why"), exceptions, and pass criteria.

`Schema.json` `WorkProfile` `Zero-Manual-Doc`

DELIVERABLE 02

## Unified Knowledge Base

Hybrid storage combining Neo4j Knowledge Graph (Equipment-Symptom-RootCause-Procedure) with PostgreSQL pgvector (dense HNSW + lexical RRF).

`Neo4j` `PGVector` `GraphRAG`

DELIVERABLE 03

## Practical LLM Assistants

Dual purpose-built agents: Station-side Work Assistant for apprentice guidance with verified citations, and RCA Agent for 5-Why root cause analysis.

`Work Assistant` `RCA Agent` `5-Why MES`

DELIVERABLE 04

## Usefulness Evaluation Report

Quantitative benchmark powered by Ragas (Faithfulness ≥ 0.85, Relevancy ≥ 0.80) and shopfloor A/B testing on novice training cycle times.

`Ragas Metric` `Golden QA 30+` `A/B Shopfloor`

**<> Suggested Tech Stack**

`LLM + RAG` `Knowledge Graph (Neo4j)` `Process/Workflow Capture` `LangChain / LlamaIndex` `Qwen3-VL (Self-Hosted)` `pgvector + FastAPI`

REFERENCE CASE · CASE THAM KHẢO

# When Knowledge Capture Turns Exploitative

"Factory Workers Are Training AI to Replace Them... Here's the Truth" · Source: youtu.be/5BoTtw8s33g

METHODOLOGY · CÁCH LÀM

## Passive & Covert Extraction

- Garment and assembly workers in India wear head-mounted cameras across entire 8-hour shifts.
- Recording complete fine-motor hand gestures — capturing egocentric data impossible to recreate in AI laboratories.
- Strategic deployment: Extremely low labor cost (~$240/month) with zero cross-border AI data restrictions.

CONSEQUENCES · HỆ QUẢ ĐÁNG LO NGẠI

## Backlash & Ethical Hazards

- Rapid expansion beyond manufacturing: construction, surgery, logistics — all manual skilled trades.
- Cognitive professions face identical vulnerability (e.g., Scale AI hiring software engineers to evaluate and debug code models).
- Low-wage workers inadvertently train their automated replacements without consent, attribution, or gain-sharing.

> *"Macro Question: If AI displaces workers entirely, who will retain the purchasing power to consume manufactured goods?"*

LESSONS LEARNED · BÀI HỌC RÚT RA

# Two Opposing Approaches: Replacement vs. Empowerment

The decisive divide between failure due to worker resistance and enduring success through collaborative trust.

EXTRACT & REPLACE (CASE THAM KHẢO)

## Extraction & Replacement

Exploitation and worker displacement

PURPOSE · MỤC ĐÍCH

Replace human craftsmen through total automation

EXPERT ROLE · VAI TRÒ CHUYÊN GIA

Passive surveillance subject with zero data ownership

INCENTIVES · KHUYẾN KHÍCH

None — flat base wage, zero share in AI productivity gains

OUTCOME · KẾT QUẢ

Erosion of trust, worker alienation, and active resistance

CAPTURE & EMPOWER (MENTORMIND)

## Capture & Empowerment

Preserving know-how & elevating artisans

PURPOSE · MỤC ĐÍCH

Retain, scale, and celebrate veteran master know-how

EXPERT ROLE · VAI TRÒ CHUYÊN GIA

Active co-creator, verified reviewer, and honored mentor

INCENTIVES · KHUYẾN KHÍCH

Direct recognition, bonus credits & rewards for verbalizing the "why"

OUTCOME · KẾT QUẢ

Craftsman dignity protected, knowledge preserved, smooth handoff

SOLUTION · GIẢI PHÁP

# Non-Invasive Capture Device + Verbal Explanation Incentives

Thiết bị capture + cơ chế khuyến khích giải thích bằng lời — 4-Step Operational Pipeline.

1

## Capture

Non-invasive camera & ambient sensors record technician movements at the station without interrupting cycle takt time.

2

## Explain

Artisan pauses to verbalize underlying "why" and tactile instincts → receives immediate recognition credits and rewards.

3

## AI Process

Self-hosted VLM (Qwen3-VL) + LLM pipeline converts video and spoken commentary into structured WorkProfile schemas.

4

## Store

Knowledge Graph (Neo4j) + Hybrid RAG (pgvector) store operational know-how with permanent author attribution.

**Core Differentiator · Điểm Khác Biệt Cốt Lõi**

Artisans are incentivized and rewarded for voluntarily articulating their wisdom — never covertly surveilled or replaced as in the reference case.

— ARCHITECTURE · KIẾN TRÚC

# Technical System Architecture: 5-Layer Stack

5 Technical Steps: Capture → Standardize → Knowledge Base → AI Agents → Evaluation (Kiến trúc 5 tầng hoàn chỉnh).

**1. CAPTURE** — **Multimodal Ingestion**
Station video 4–8 fps, commentary microphone, Docling OCR for drawings & SOPs
`Docling` `Camera`

**2. STANDARDIZE** — **Multimodal VLM Pipeline**
MediaPipe Hands pose-tracking + Qwen3-VL 8B (BF16) + Task Graph synthesis
`Qwen3-VL` `MediaPipe`

**3. KNOWLEDGE BASE** — **Hybrid GraphRAG + pgvector**
Neo4j Equipment-Symptom-RootCause graph + PostgreSQL pgvector RRF + Sigma MES
`Neo4j` `pgvector`

**4. AI AGENT** — **Dual AI Agents**
Station Work Assistant (citations & refusal guards) + RCA 5-Why Agent
`Assistant` `RCA Agent`

**5. EVALUATION** — **Continuous Evaluation**
Ragas empirical benchmark (Faithfulness ≥ 0.85) + Novice A/B testing
`Ragas` `A/B KPI`

MULTIMODAL INGESTION

## Layer 1: Capture & Multimodal Ingestion

Captures high-fidelity factory data without interrupting active production flows:

- Station camera: 4–8 fps video sampling optimized for rapid hand movements.
- Commentary mic: Captures spontaneous artisan verbal explanations.
- Docling / mmore: Structured OCR extraction of engineering drawings and assembly SOPs.

View High-Resolution Architecture Blueprint

— ROADMAP · LỘ TRÌNH TRIỂN KHAI

# From Capture to Production AI Assistant in 3 Months

ROADMAP 3 THÁNG — Execution plan strictly aligned with Denso Hackathon evaluation gates.

MONTH 01

### Capture & Standardization

- ✓ Establish standardized WorkProfile template (JSON Schema).
- ✓ Deploy Docling OCR pipeline for drawings & assembly SOPs.
- ✓ Ingest pilot video on Kitting & Packing station (Asian Alliance / Denso).

MONTH 02

### Knowledge Base & AI Agents

- ✓ Deploy Neo4j Knowledge Graph + PostgreSQL pgvector (RRF).
- ✓ Ship station-side Work Assistant with strict citation verification.
- ✓ Develop RCA Agent with 5-Why reasoning linked to Mock Sigma MES.

MONTH 03

### Evaluation & Pilot Rollout

- ✓ Run Ragas benchmark across Golden Set (30+ validation queries).
- ✓ A/B testing on novice technicians learning station workflows.
- ✓ Latency optimization and packaging final Usefulness Evaluation Report.

## "Preserve Master Know-How — Empower Human Artisans, Never Replace Them."

"Giữ lại tri thức — Trao quyền cho con người, không thay thế." MentorMind stands for human-centric industrial AI: celebrating master craftsmanship, passing on tacit expertise, and empowering the next generation of technicians.

DENSO HACKATHON 2026 · THANK YOU & Q&A